

SUSAN GHIMIRE

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SUMMARY

Machine Learning Engineer with hands-on experience shipping MLOps pipelines and LLM-powered features, backed by deep computer vision work across surgical AI, 3D reconstruction, and model optimization.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL

ML Frameworks & Libraries: PyTorch, TensorFlow/Keras, Scikit-Learn, NumPy, SciPy, Pandas, Matplotlib, OpenCV, Open3D

Model Optimization: NVIDIA ModelOpt, TensorRT, ONNX, CoreML, Quantization, Knowledge Distillation

Inference & Serving: NVIDIA Triton Inference Server, FastAPI, Pydantic, gRPC

Agentic AI & GenAI: Google ADK, LLM, RAG, Diffusion Models, GroundingDINO, SAM

Experimentation & MLOps: Hydra, MLflow, AWS SageMaker, Docker, GitHub Actions, CI/CD

Computer Vision & 3D: NeRF Studio, Gaussian Splatting, Photogrammetry, Structure-from-Motion, MMDetection, YOLO, Vision Transformers

Annotation & Labeling: Label Studio, CVAT

Hardware & Embedded: Arduino, Raspberry Pi, ROS2

WORK EXPERIENCE

Machine Learning Engineer (Consultant)

Dec 2025 - Feb 2026

AI Inference Acceleration Pipeline

Remote, Ireland

- Pruned, sparsified, and quantized vision models, hitting up to 7x faster inference on NVIDIA GPUs within 2% of baseline accuracy.
- Recovered accuracy from 80.26% to 82.88% by grafting in a teacher-student distillation stage post-optimization, at half the original parameter count.
- Ported ZipVoice (123M params) from PyTorch through ONNX to TensorRT with FP16 and dynamic shapes, served via Triton at 2.84x faster inference without quality loss.

Machine Learning Engineer

Jun 2022 - Present

Fusemachines Nepal Pvt. Ltd.

Nepal

Database Schema Mapping

- Built a 3-stage AI mapping engine that automatically maps source data schemas to a canonical model via an API-driven pipeline, hitting 70% mapping accuracy out of the box.
- Wired in a YAML-based config layer so model parameters, feature weights, and matching strategies can be swapped without changing code.

Demand Forecasting

- Built a forecasting engine that models how price changes ripple through sales and revenue, giving pricing teams a data-backed view before committing to changes.
- Layered in post-forecast calculations for MoM change, cumulative YTD performance, and baseline impact deltas, so analysts get business-ready numbers without extra processing.
- Plugged in an LLM to let users ask questions about forecast results in plain language, cutting the gap between model output and business decisions.

AI-Assisted Surgery

- Deployed an AWS MLOps pipeline for saliency detection in surgical videos using SageMaker for inference and MLflow for experiment tracking, versioning, and automated deployments.
- Built an annotation tool for salient region detection in surgical videos, reaching over 90% annotation accuracy across a 40k+ image dataset covering object detection and occlusion scenarios.
- Cut novel view synthesis rendering from hours down to 7-20 minutes through algorithmic optimizations, and trimmed the multi-camera surgical capture rig from 30 cameras to 6-9 without losing scene coverage.

- Reconstructed static and dynamic 3D surgical scenes using NeRF and Gaussian Splatting and built an interactive tool for blending Gaussian splats with 3D meshes with playback and navigation controls for 4D visualization.
- Shipped prompt-based surgical instrument segmentation using GroundingDINO and SAM, and added tracking via MMDetection, as part of a markerless surgical tracking system for a US-based orthopedic healthcare client.

Autonomous Greenhouse Robot

- Worked on computer vision systems for an autonomous greenhouse robot, handling pollination and plant-lowering tasks while mapping the greenhouse environment and applying Object detection and recognition in 3D.

Table Detection and Document Analysis

- Fine-tuned Cascade TabNet and TabStructNet for table detection and structure recognition on financial receipts, covering dataset generation, augmentation, and labeling in Label Studio and CVAT.

Teaching Assistant

May 2024 - Oct 2024

Fusemachines Nepal Pvt. Ltd.

Nepal

- Supervised 11 capstone ML projects across CV, NLP, and forecasting domains for a cohort of 30+ fellows, keeping 100% of milestones on track across six modules.
- Ran weekly TA sessions covering core ML concepts and hands-on labs in computer vision, NLP, and forecasting, with applications in education and agriculture.

EDUCATION

Bachelor's in Computer Engineering

Nov 2017 - Apr 2022

Tribhuvan University

Nepal

PUBLICATIONS

CatDox: Document Categorization and Extraction

Apr 2025

- Built CatDox, a document processing pipeline for finance, chaining ResNet classification, DiT layout detection, and Devanagari-tuned OCR to pull key-value fields from scanned Nepali and English documents, including passports, PAN cards, and account forms, presented in GIBL Hackathon and awarded in extremely useful category.
- Applied SIFT-based alignment before inference and a ByT5 correction layer after OCR, so the pipeline could handle skewed scans and messy character errors without manual cleanup.

Insights on Simulation and Modelling

Dec 2023

- Co-authored simulation and modeling guidebook with a professor tailored for Bachelor of Computer Engineering students, adopted for use in an active course.

FELLOWSHIP

Microdegree in ML, DL, CV, NLP

Mar 2022 - Jan 2024

Fusemachines AI Fellowship Program 2022

Nepal

- Completed a structured microdegree covering ML, deep learning, computer vision, and NLP, including supervised/unsupervised learning, CNNs, RNNs, transformers, object detection, 3D reconstruction, and sequence modeling.

LEADERSHIP & MENTORSHIP

President

Mar 2021 - May 2022

Robotics Club, Kathmandu Engineering College

Nepal

- Built terrestrial robots using ROS, Raspberry Pi, GPS, and radio controls, and developed a quadcopter delivery system on Pixhawk.
- Led workshops on IoT and robotics, designed smart home automation systems, and coordinated multiple hackathons and technical events.